HISHAM MOHAMED

AI / ML Engineer · Computer Vision · NLP · Deep Learning

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PROFESSIONAL SUMMARY

AI/ML Engineer and Artificial Intelligence graduate specializing in computer vision, deep learning, and NLP. Experienced across the full model lifecycle — data preprocessing, training, evaluation, and FastAPI deployment — using Python, PyTorch, Hugging Face Transformers, and YOLO. Skilled at turning research models into production-ready inference services with structured, usable outputs.

TECHNICAL SKILLS

Languages & Data: Python, SQL, C++, Pandas, NumPy, Data Cleaning, EDA, Feature Engineering, Data Visualization

Machine Learning: Scikit-learn, Logistic Regression, SVM, KNN, Decision Trees, Random Forest, Clustering, Recommendation Systems, Model Evaluation

Deep Learning & CV: PyTorch, PyTorch Lightning, Hugging Face Transformers, Donut, YOLO Object Detection

NLP & Generative AI: NLTK, spaCy, TF-IDF, Word2Vec, NER, Prompt Engineering, LLM Fundamentals

Deployment & Tools: FastAPI, REST APIs, Git, GitHub, Jupyter Notebook, Google Colab

PROJECTS

VisionSafe360 — AI Industrial Safety Monitoring Platform *Computer Vision · YOLO*

- Developed a real-time YOLO monitoring platform detecting PPE non-compliance, worker falls, forklift-worker proximity hazards, and restricted-zone violations across industrial environments.
- Integrated four independent detection modules into a unified workflow with a live dashboard delivering instant alerts for rapid hazard response.

OCR-Free Invoice Information Extraction — Donut Transformer *PyTorch · PyTorch Lightning · Hugging Face · FastAPI*

- Curated and split 973 SROIE receipt images (778 / 97 / 98 train-validation-test) and fine-tuned the Donut transformer for OCR-free extraction of company, date, address, and total fields.
- Built a mixed-precision training pipeline with custom task tokens, gradient accumulation, early stopping, and checkpointing, reaching a best validation loss of 0.7747.
- Deployed the model via FastAPI supporting single, batch, and URL-based extraction, returning structured JSON through a drag-and-drop interface.

Olist E-Commerce Recommendation System *Python · Scikit-learn · TF-IDF*

- Built a content-based product recommendation engine using TF-IDF, cosine similarity, and Nearest Neighbors on a real-world e-commerce catalog.
- Evaluated ranking quality with Precision@K, Recall@K, and NDCG@K to validate recommendation relevance.

EDUCATION

B.Sc. in Artificial Intelligence — Egyptian Russian University (ERU) Jul 2026

Relevant Coursework: Machine Learning, Deep Learning, Data Science, Computer Vision, Natural Language Processing, Programming

CERTIFICATIONS & TRAINING

Machine Learning Engineer Track — Digital Egypt Pioneers Initiative (DEPI), MCIT Nov 2024 – May 2025

Building LLM Applications with Prompt Engineering — NVIDIA Sep 2025

Big Data Analysis — ITIDA & National Telecommunication Institute (120 hrs, Score: 90%) Aug 2025

Deep Learning Course — Egyptian Russian University Jan – Feb 2026

LANGUAGES

Arabic (Native) · English (Professional Working Proficiency)